WORKSHEET

Chapter 10: Polls and levels of confidence

1 A poll conducted after a particularly brutal murder was based on 1500 mobile telephone interviews conducted by trained interviewers. Mobile telephone numbers were selected at random. According to the poll, the re-introduction of the death penalty was supported by 52% of those interviewed while 48% were against any change. The margin of error was stated as 3%.

Based on these results, a news report ran the headline was 'Overwhelming support for the death penalty'.

a What does the 95% confidence level for two-party support mean?

b Based only on this poll, is the headline justified?

c Does the method of polling introduce any bias?

2 The average height of pine seedlings is 124 cm with a standard deviation of 44 cm from the measurement of 100 seedlings in a plantation. Assume the height is randomly distributed, so the 100 measurements can count as a random sample.

a What is the standard deviation of the sample mean?

b What is the probability that the average of the next 10 seedlings will exceed 150 cm?

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c The tallest seedling in the plantation was measured to be 234 cm. Does this cast any doubt on the assumption that the rainfall is randomly distributed?

3 The average annual rainfall in Tully is 4100 mm, with a standard deviation of 1550 mm. The highest official rainfall recorded was 7900 mm in 1950. The average is based on 50 years of rainfall records.

a What is the standard deviation of the mean?

b What is the probability that the average annual rainfall over the next 8 years will exceed 3500 mm?

4 A research company published the results of its latest poll and stated that the results are within a margin of error of 3%.

a What would the margin of error be at the 90% level of confidence?

b What would the margin of error be at the 99% level of confidence?

c CAS Find the margin of error at the 99.9% level of confidence.

5 The ABS conducted 2500 interviews of randomly selected people over 18 to compile data on the average time spent watching TV. According to the survey, the average time spent watching TV was 24.5 hours per week. If the variance of the hours TV was watched was 60, what is the 90% confidence interval?

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Answers

1 **a** Support for the death penalty is between 49% and 55%, while against the death penalty is between 45% and 51%.

b At the 95% confidence level, support for the death penalty could be as low as 49% with those against the death penalty running as high as 51%. Therefore the headline is not justified.

c There is bias towards people who have mobile telephones. People with only landlines are excluded, so it is likely that there is a bias towards younger people.

2 **a** 4.4 cm

b 0.031

c No, because if it is random a very tall seedling is possible.

3 **a** 212 mm

b 87%

4 **a** 2.52%

b 3.94%

c 5.04%

5 The 90% confidence interval for the time watching TV was about [24.25, 24.75].

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